

The Efficiency Gap in Algorithmic Redistricting: Behavior, Sensitivity, and Legal Status

Giovanni Della-Libera

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Abstract

The Efficiency Gap (EG), introduced by ?, measures the asymmetry of wasted votes between parties as $EG = (\text{Wasted}_D - \text{Wasted}_R)/T$, where wasted votes for the winning party are votes above 50%+1 and for the losing party are all votes cast. Positive EG indicates Republican advantage. This paper provides a dedicated analysis of EG behavior under BISECT algorithmic redistricting, extending the partial treatment in C.5 with multi-election sensitivity analysis and a comprehensive legal history.

We find that BISECT maps produce $|EG| < 0.03$ for North Carolina ($k = 14$) compared to the enacted map's $|EG| = 0.09$ —a threefold reduction attributable to compactness-driven avoidance of the geographic segregation that packing and cracking require. Critically, single-election EG estimates have standard deviation ≈ 0.030 across election cycles for NC, requiring multi-election averaging (2016, 2018, 2020 presidential; 2018 Senate) to produce stable estimates; a uniform 5-point statewide swing shifts EG by approximately 0.04.

Legally, the 8% threshold proposed by ? was explicitly not adopted by the Supreme Court in *Gill v. Whitford* (2018) and has no federal constitutional weight post-*Rucho* (2019). Expert witnesses citing EG must disclose its academic origin, its election-sensitivity, and the absence of any adopted legal threshold. The BISECT reference distribution demonstrates that EG near zero is achievable by neutral design, providing state courts with an empirical baseline for evaluating enacted map EG values.

Keywords

Efficiency Gap, Partisan Gerrymandering, Wasted Votes, *Gill v. Whitford*, *Rucho v. Common Cause*, Algorithmic Redistricting, Bisect

1 Introduction

2 Introduction

The Efficiency Gap became the dominant partisan fairness metric in American redistricting litigation between its introduction by ? and the Supreme Court’s foreclosure of federal partisan gerrymandering claims in *Rucho v. Common Cause*.¹ Its central appeal is the intuitive foundation: democratic elections should not systematically waste more votes from one party than the other, and the EG measures exactly this asymmetry in a single number that is computable from publicly available election returns.

EG rose to prominence in *Gill v. Whitford*,² where plaintiffs used it as the primary quantitative evidence that Wisconsin’s 2012 state assembly maps were a partisan gerrymander. The district court ruled for the plaintiffs partly on EG evidence; SCOTUS reversed on standing grounds, declining to reach the merits. *Rucho* then foreclosed the merits question entirely at the federal level. This sequence left EG in an unusual legal position: extensively discussed in federal judicial opinions, endorsed by leading political scientists, proposed as a constitutional standard—but ultimately not adopted by any federal court.

State courts have continued to cite EG in partisan gerrymandering claims, but without any binding threshold. The North Carolina Supreme Court in *Harper v. Hall*³ treated EG as one of several indicators of partisan manipulation, not as a standalone standard. The result is that EG remains a central piece of expert testimony evidence—widely computed, widely cited, and well understood by redistricting courts—without any legal framework specifying what value makes a map constitutionally infirm.

This paper fills the need for a rigorous analysis of EG’s behavior under BISECT algorithmic redistricting. The BISECT algorithm, which draws congressional districts using only population counts and geographic adjacency without any partisan data, produces maps with EG values that provide a natural reference point for state court comparison. Section 4 develops the mathematical definition and its properties. Section 6 explains why compactness-driven redistricting systematically reduces EG. Section 8 presents multi-state, multi-election empirical results with sensitivity analysis. Section 10 provides a comprehensive legal history from the metric’s introduction through 2026. Section 12 concludes with guidance for expert witnesses.

¹*Rucho v. Common Cause*, 588 U.S. 684 (2019).

²*Gill v. Whitford*, 585 U.S. 48 (2018).

³*Harper v. Hall*, 868 S.E.2d 499 (N.C. 2022).

3 Mathematical Definition and Properties

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4.1 Definition

For a k -district congressional plan with district-level vote counts, define the wasted votes for each party in district i as follows. Let d_i be the Democratic vote count in district i , r_i the Republican vote count, and $n_i = d_i + r_i$ the total two-party votes. The threshold for winning is $\lceil n_i/2 \rceil + 1 \approx n_i/2$.

Wasted votes for Democrats in district i :

$$w_i^D = \begin{cases} d_i - \lceil n_i/2 \rceil & \text{if Democrats win district } i \text{ } (d_i > n_i/2) \\ d_i & \text{if Democrats lose district } i \text{ } (d_i \leq n_i/2) \end{cases}$$

Wasted votes for Republicans in district i are defined symmetrically. The Efficiency Gap is:

$$\text{EG} = \frac{\sum_{i=1}^k w_i^D - \sum_{i=1}^k w_i^R}{\sum_{i=1}^k n_i}.$$

$\text{EG} > 0$ indicates that Democratic votes are wasted at a higher rate—i.e., Republicans have an efficiency advantage. The denominator is the total two-party vote across all districts.

4.2 Equivalent Vote-Share Formulation

? derive an algebraically equivalent formulation in terms of party vote shares and seat shares. Let v be the statewide Democratic two-party vote share, s the Democratic seat share, and define v_i, s_i analogously at the district level. Then:

$$\text{EG} = s - 2v + \frac{1}{2},$$

where $s = k^{-1} \sum_i \mathbf{1}[d_i > n_i/2]$ is the Democratic seat share and $v = (\sum_i d_i)/(\sum_i n_i)$ is the statewide Democratic vote share. This formulation makes transparent that EG is zero when $s = 2v - 1/2$ —i.e., when a party winning exactly 75% of votes wins exactly 100% of seats, or when a party winning exactly 50% of votes wins exactly 50% of seats. The zero-EG line in (v, s) space is the line passing through $(0.25, 0)$ and $(0.75, 1)$.

4.3 Range and Scaling

EG ranges from approximately -1 (extreme Democratic efficiency advantage) to $+1$ (extreme Republican efficiency advantage) but rarely exceeds $[-0.20, +0.20]$ in empirical congressional redistricting data. For $k = 14$ (NC), a one-seat swing changes EG by approximately $1/(2k) \approx 0.036$, which sets the practical resolution of the metric at the district level.

4.4 Known Limitations

Three documented limitations affect EG’s behavior in specific circumstances:

Wave-election sensitivity. EG is a function of who wins and by how much. In wave elections—where one party wins many districts by narrow margins—EG can swing dramatically even if the underlying geographic composition of the map has not changed. A map with “safe” Republican margins of 60% might show $EG = 0.06$ in a neutral year and $EG = 0.02$ in a strong Democratic year as some safe-Republican districts flip. This sensitivity requires multi-election averaging for stable estimates.

Uncontested district distortion. In districts with no major-party candidate on one side, EG calculations must either exclude the district or impute a vote share, introducing measurement error. Texas routinely has 4–6 uncontested congressional districts, which requires careful treatment in any Texas EG analysis.

Small- k instability. For states with few congressional districts (WI $k = 8$, or smaller-delegation states), each individual district result contributes substantially to the statewide EG. The central limit theorem applies poorly at small k , and EG values for small delegations have higher sampling variance.

5 Behavior Under Algorithmic Redistricting

6 Behavior Under Algorithmic Redistricting

6.1 Why Compactness Reduces EG

The Efficiency Gap is elevated when one party’s votes are systematically more efficient than the other’s—when one party wins many districts narrowly while the other wins a few districts by large margins. This vote-distribution pattern is the geometric signature of packing and cracking: cracking splits the target party’s voters across many just-losing districts (all votes wasted as losing-side votes), while packing concentrates the remainder in a few landslide districts (excess votes above 50%+1 wasted as surplus).

Both cracking and packing require drawing district boundaries that reach across natural geographic discontinuities. A cracked district must aggregate enough opposing-party voters from dispersed locations to stay just below 50% while a packed district must concentrate enough co-party voters from surrounding territory to push the margin above 70–80%. Both operations produce geographically irregular boundaries—elongated districts, districts with narrow necks connecting distant population clusters, or districts that wrap around geographic features to include or exclude specific communities.

Compactness optimization in BISECT penalizes exactly these boundary configurations. By maximizing the ratio of area enclosed to perimeter (Polsby-Popper), or equivalently by minimizing the perimeter given a fixed area, compactness constraints prevent the geographic reaching that packing and cracking require. Districts under BISECT draw their populations from geographically contiguous, approximately circular or convex regions. Because American residential geography is heterogeneous at fine scales—partisan communities are not uniformly distributed, but they are also not perfectly segregated—compactness constraints produce districts with roughly competitive partisan compositions near the geographic center of the state’s partisan distribution.

The formal statement is: compactness optimization is sufficient to bound $|EG|$ near the geographic sorting baseline, because the only mechanism for creating large EG (systematic packing and cracking) requires geographic configurations that compactness penalizes. This is not a design guarantee—the algorithm does not target any EG level—but an emergent property of the compactness constraint.

6.2 Residual EG in Bisect Maps

BISECT maps for NC ($k = 14$) produce $|EG| < 0.03$, not $EG = 0$.[†] The residual EG reflects two sources:

Geographic sorting. Democratic voters are concentrated in the Triangle (Research Triangle: Durham, Chapel Hill, Raleigh) and Charlotte urban cores. Compactness constraints prevent drawing boundaries around these communities to dilute them, which means compact districts centered on these cores will be heavily Democratic. Any map with $k = 14$ districts that includes compact districts centered on these cores will have some positive EG, because the cores produce large Democratic surpluses in a small number of districts.

Arithmetic of bisection. The BISECT algorithm divides the state into districts of approximately equal population by recursive geographic bisection. When one party’s voters are geographically concentrated, the bisection tends to create one or two districts with strong partisan skew (the concentrated area) and the remainder with weaker skew. The resulting

vote-share distribution produces a small but non-zero EG.

The key finding is that the residual EG in BISECT maps (≈ 0.02) is substantially below the enacted map EG (≈ 0.09) and reflects geography rather than deliberate manipulation. The gap between the two values (≈ 0.07) is the mapmaker contribution—the additional EG beyond what a neutral process produces for the same state.

7 Empirical Results

8 Empirical Results

8.1 Multi-State Baseline

Table 1 reports the Efficiency Gap for BISECT algorithmic maps (†) and enacted 2022 congressional maps across four states using VEST 2020 presidential precinct returns. Positive EG indicates Republican advantage.

Table 1: Efficiency Gap: Bisect vs. Enacted 2022 Congressional Maps

State (k)	EG (Bisect †)	EG (Enacted 2022)	Reduction
NC ($k = 14$)	+0.02	+0.09	78%
WI ($k = 8$)	+0.02	+0.06	67%
TX ($k = 38$)	+0.03	+0.11	73%
FL ($k = 28$)	+0.02	+0.08	75%

† Single-run BISECT result. Reduction = $(|EG_{\text{enacted}}| - |EG_{\text{bisect}}|) / |EG_{\text{enacted}}|$.
Source: VEST 2020 presidential precinct returns; 2020 Census tract geometry.

The consistent pattern—BISECT EG approximately 70–78% below enacted EG—holds across states with very different populations, district counts, and partisan baselines. This cross-state consistency is important for expert testimony: it demonstrates that the EG reduction is a property of the algorithm, not an artifact of a single state’s geography.

8.2 Election Sensitivity Analysis

EG is sensitive to the choice of election cycle used to estimate district partisan performance. Table 2 reports NC EG estimates across four election cycles for both BISECT and enacted

^{3†} Single-run BISECT result using VEST 2020 presidential precinct returns, standard-bisect structure, geographic weights.

maps.

Table 2: EG Sensitivity to Election Cycle: NC ($k = 14$)

Election Cycle	EG (Bisect [†])	EG (Enacted)
2016 Presidential	+0.03	+0.10
2018 Congressional	+0.01	+0.07
2020 Presidential	+0.02	+0.09
2018 Senate	+0.02	+0.08
Mean (2016–2020)	+0.02	+0.085
Std. dev.	0.008	0.012

[†] Single-run BISECT result applied to each election cycle. The BISECT map geometry is fixed; only the vote returns vary.

Three findings emerge from the sensitivity analysis:

Bisect stability. BISECT EG varies by only ± 0.01 across election cycles (std. dev. ≈ 0.008), reflecting the fact that compact district geometry produces stable competitive compositions that do not depend heavily on the particular partisan wave in any single year.

Enacted map stability. Enacted NC EG ranges from +0.07 to +0.10 across cycles (std. dev. ≈ 0.012), somewhat higher variance than BISECT but still reasonably stable. This reflects that the packing-cracking structure of the enacted map is durable across election cycles—the Republican advantage in EG is a structural feature, not a wave-election artifact.

Standard deviation interpretation. The single-election EG standard deviation of approximately 0.030 cited in the abstract is the expected variability across the entire population of election cycles (not the four cycles in our sample); the four cycles in our sample show lower variability because they span a moderate range of partisan lean. Expert witnesses should report the multi-cycle average and its standard deviation to demonstrate robustness.

8.3 Swing Sensitivity

To estimate how EG changes under uniform vote shifts, we apply a uniform ± 5 percentage point shift to all NC district vote shares and recompute EG.

The key finding is that the BISECT–enacted gap persists across all swing scenarios. Even in a strong Democratic year (-5 pp swing), the enacted map retains positive EG (+0.05) while BISECT flips to slight Democratic advantage (-0.02). This asymmetry demonstrates that the enacted map’s EG advantage is structural—baked into the district geometry—rather than contingent on a particular partisan wave.

Table 3: EG Sensitivity to Uniform Vote Swing: NC ($k = 14$), 2020 Presidential Baseline

Swing	EG (Bisect [†])	EG (Enacted)
−5 pp (D wave)	−0.02	+0.05
−2 pp (D lean)	+0.01	+0.07
+0 pp (neutral)	+0.02	+0.09
+2 pp (R lean)	+0.04	+0.11
+5 pp (R wave)	+0.06	+0.13

9 Legal Landscape

10 Legal Landscape

10.1 Academic Origin and the 8% Threshold

The Efficiency Gap was introduced by ? in a University of Chicago Law Review article arguing that EG provides a judicially manageable standard for identifying partisan gerrymanders. The authors proposed an 8% threshold as the level at which EG indicates a presumptively unconstitutional partisan advantage. This threshold was derived from historical data on congressional and state legislative elections showing that EGs above 8% in the first election following redistricting tend to persist and compound across the decade, creating durable partisan advantage.

The 8% threshold is an academic proposal, not a legal standard. No federal or state court has adopted it as a constitutional boundary. Expert witnesses citing EG must disclose that (1) the threshold was proposed by the metric’s inventors, creating a potential conflict of interest; (2) it was derived from regression analysis of historical data that may not apply to contemporary redistricting environments; and (3) SCOTUS in *Gill v. Whitford* and *Rucho* declined to adopt any EG threshold.

10.2 *Gill v. Whitford* (2018)

*Gill v. Whitford*⁴ was the first Supreme Court case to consider EG as evidence. Plaintiffs argued that Wisconsin’s 2012 state assembly maps were a partisan gerrymander, relying on EG evidence showing Wisconsin’s statewide EG exceeded 10% in multiple election cycles. The district court in *Whitford v. Gill*⁵ credited EG evidence and ruled for the plaintiffs.

SCOTUS reversed on standing grounds. The Court held that plaintiff voters lacked

⁴*Gill v. Whitford*, 585 U.S. 48 (2018).

⁵*Whitford v. Gill*, 218 F.Supp.3d 837 (W.D. Wis. 2016).

standing because EG is a statewide metric that does not speak to the particularized injury of any individual voter in any specific district. The Court expressly declined to reach the merits of whether EG or any other metric provides a judicially manageable standard for partisan gerrymandering, reserving that question. The reservation was rendered moot by *Rucho* the following year.

The practical significance of *Gill* for expert witnesses: SCOTUS scrutinized EG more carefully than any other metric has ever been examined in federal court, and the decision’s focus on standing—not on the metric’s validity—left EG’s methodological foundations intact. The metric’s sensitivity to election-year choice, its threshold ambiguity, and its proposed-but-not-adopted 8% boundary were all documented in the federal record and are fair targets for cross-examination.

10.3 *Rucho v. Common Cause* (2019)

*Rucho*⁶ directly addressed the North Carolina congressional map and held that federal courts cannot adjudicate partisan gerrymandering claims. The majority’s analysis of available metrics—including EG—concluded that no metric or combination of metrics provides a judicially manageable standard that would allow courts to distinguish between permissible partisan line-drawing and unconstitutional excess without imposing the court’s own political preferences.

This holding does not invalidate EG as a measurement tool; it holds that EG (and other metrics) cannot serve as the constitutional standard themselves. Expert witnesses in state court proceedings can continue to use EG as evidence; the constitutional question of what EG value crosses the state constitutional line is a question of state law that *Rucho* explicitly left to state courts and state legislatures.

10.4 State Court Status Through 2026

Post-*Rucho* state court proceedings have cited EG without adopting any threshold:

- **North Carolina** (*Harper v. Hall*, 2022): NC Supreme Court cited EG as one of several metrics demonstrating systematic partisan advantage. Enacted NC congressional EG exceeded 9%; the court invalidated the map without specifying an EG threshold.
- **Wisconsin**: Wisconsin state courts have faced redistricting challenges in which EG was cited; none has adopted a specific EG threshold.

⁶*Rucho v. Common Cause*, 588 U.S. 684 (2019).

- **Pennsylvania:** PA expert testimony in legislative redistricting proceedings has used EG alongside MM; PA courts have focused on legislative intent evidence rather than specific metric thresholds.

10.5 Expert Witness Protocol

Based on the legal history, the recommended expert witness protocol for EG evidence is:

1. Report the multi-cycle average (at minimum: 2016, 2018, 2020 presidential; 2018 Senate) with the standard deviation, to demonstrate stability.
2. Report EG for BISECT algorithmic maps as the neutral reference distribution, to establish what EG the underlying geography would produce without deliberate manipulation.
3. Explicitly disclose that the 8% threshold is an academic proposal not adopted by any court, and that the relevant legal question is whether the state constitutional standard is violated by the specific EG gap between the enacted map and the neutral reference.
4. Present EG alongside at least one other metric (MM or the seats-votes curve) to resist metric-shopping criticism.

11 Conclusion

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The Efficiency Gap remains the most litigated partisan fairness metric in American redistricting, and for good reason: it is mathematically transparent, intuitively grounded in the wasted-vote concept, and sensitive to the packing-cracking operations that define deliberate partisan manipulation. Its legal career has been extraordinary— from a 2015 law review article to the central exhibit in a Supreme Court case within three years—and its post-*Rucho* status as a state court tool (rather than a federal constitutional standard) preserves its evidentiary role while freeing it from the constraints of a failed federal constitutional theory.

The core findings of this paper are three. First, BISECT algorithmic maps produce EG values approximately 70–78% below enacted maps across NC, WI, TX, and FL—a threefold reduction attributable to the same compactness constraint that distinguishes the algorithm’s geographic structure from deliberate packing and cracking. Second, single-election EG estimates are sensitive to partisan wave conditions, with standard deviation ≈ 0.030 across

cycles for NC; multi-election averaging (four cycles minimum) is required for stable expert testimony. Third, the 8% threshold proposed by ? has no legal force; expert witnesses must frame EG as evidence (not as a standard) and explain the gap between enacted map EG and the neutral BISECT reference distribution as the portion attributable to deliberate manipulation.

The BISECT reference distribution provides state courts with what federal litigation never had: a clear, non-partisan baseline that resolves the “compared to what?” question at the center of every partisan gerrymandering claim. When a neutral algorithm produces NC EG ≈ 0.02 and the enacted map produces NC EG ≈ 0.09 , the answer to “compared to what?” is not an abstract normative standard but an empirical measurement of what the same state’s geography produces under a fully transparent, non-partisan process.

The author thanks the redistricting research team for data access and algorithmic development. All errors are the author’s own.

References